Pixel vs Symbolic Observations for PPO

Super Mario Bros Reinforcement Learning

Lucas Martim Sergio Contente Leonardo Falabella
Gabriel Corsi Lara Polachini

CSC-52081 — Reinforcement Learning

March 17, 2026

Outline

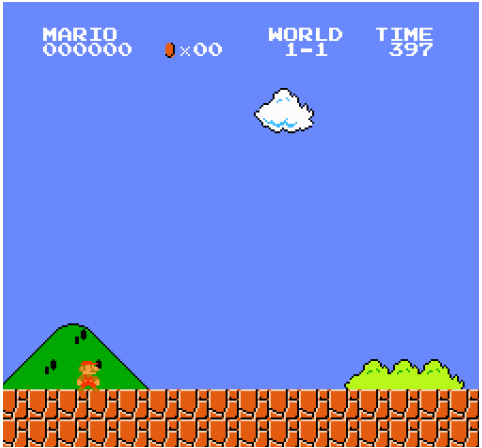
1. Motivation
2. Project Overview
3. Methodology
4. Results
5. Discussion & Conclusion

Motivation

Why Mario?

- Classic RL benchmark
- Semi-Sparse rewards, precise control
- High-dimensional pixel observations

Our question:
Can *symbolic* (RAM-based) state + PPO beat pixel-based DQN?



Project Overview

Experiments, in order:

1. DQN baseline (symbolic obs.)
2. Pixel PPO (CNN)
3. Symbolic PPO (RAM grid)
4. Transfer: World 1-1 $\rightarrow$ 1-2
5. Multi-task: From 1-1 to 1-4 jointly
6. Random start positions

Two observation types

Pixel	Raw frames + CNN
Symbolic	NES RAM $\rightarrow$ tile grid

RAM = structured & compact,
but requires domain knowledge.

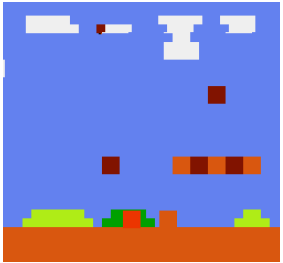
Methodology: Observations

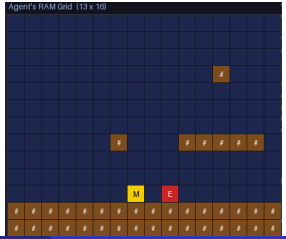
Pixel

- 84×84 grayscale, 4 stacked frames
- Nature CNN $\rightarrow$ FC 512
- **78 FPS** on Colab T4

Symbolic (RAM)

- 13×16 tile grid, 4 stacked
- Encodes: terrain, enemies, Mario
- MLP [512, 512]
- **500 FPS** on CPU





Methodology: Reward Shaping

$$r \leftarrow \frac{1}{10}\left[r_{\text{env}} + \frac{\Delta\text{score}}{40} + 50 \cdot \mathbf{1}[\text{flag}] - 50 \cdot \mathbf{1}[\text{death}]\right]$$

- r_{env}: horizontal progress
- $\Delta\text{score}/40$: coins, kills, blocks
- ± 50: flag bonus / death penalty
- $\div 10$: normalises reward scale

Results: World 1-1 Summary

Symbolic DQN — 0% flag

Never escapes the first obstacle.

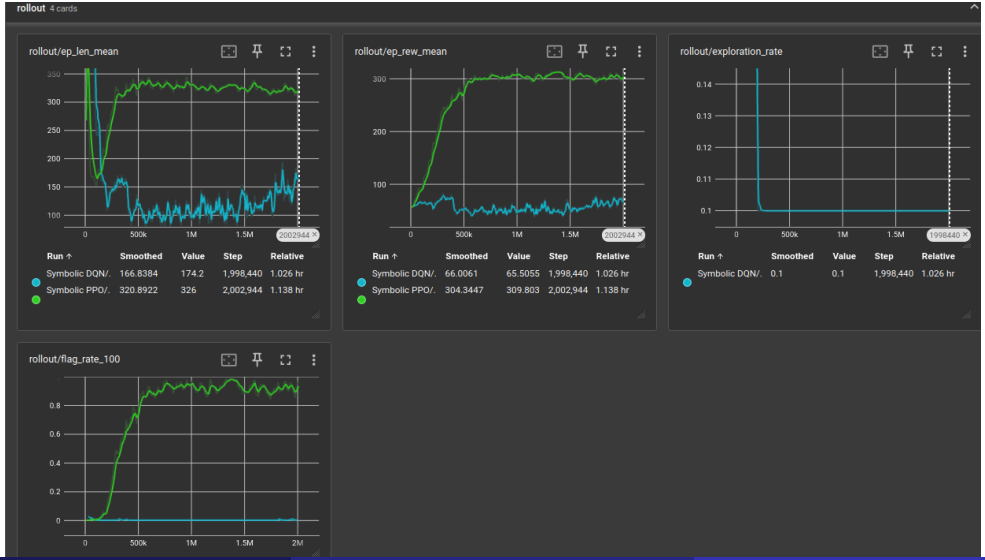
Pixel PPO — >80% flag

Mean reward ≈ 285. Slow: 78 FPS on GPU.

Symbolic PPO — 93% flag (peak 97%)

Mean reward ≈ 304. Fast: 500 FPS on CPU.

Results: World 1-1 Summary (Training Curves)



Results: Transfer Learning 1-1 $\rightarrow$ 1-2

Setup

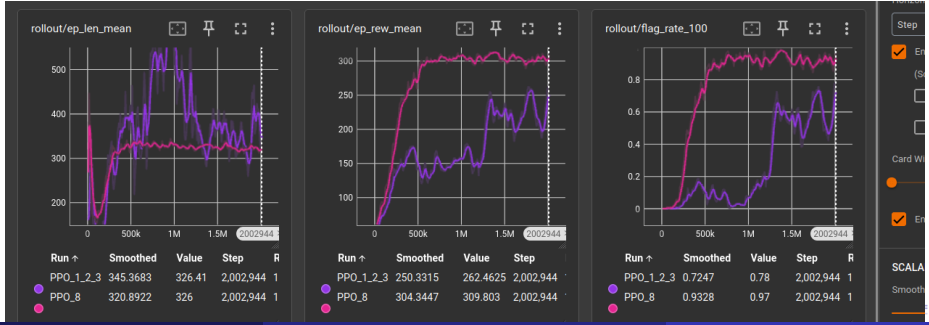
Train on 1-1 $\rightarrow$ fine-tune on 1-2
(lr $= 10^{-5}$, weights unfrozen)

World 1-2: **78% flag rate**

Converges faster than from scratch.

World 1-1: **0% flag rate**

Complete performance collapse.



Results: Random-Start Curriculum (World 1-1)

Problem: fixed start $\Rightarrow$ agent memorises one trajectory.

Two-layer solution:

1. **Random-start wrapper** — at each reset, execute $n \sim \mathcal{U}[0, N_{\max}]$ forward actions (right / right+A / right+B / right+A+B) Mario *walks* into the level, not teleported.
2. **Curriculum callback** — linearly anneal $N_{\max} : 0 \to 100$ over training (every 1 k steps). Easy start first, then harder as the agent improves.

Design separation

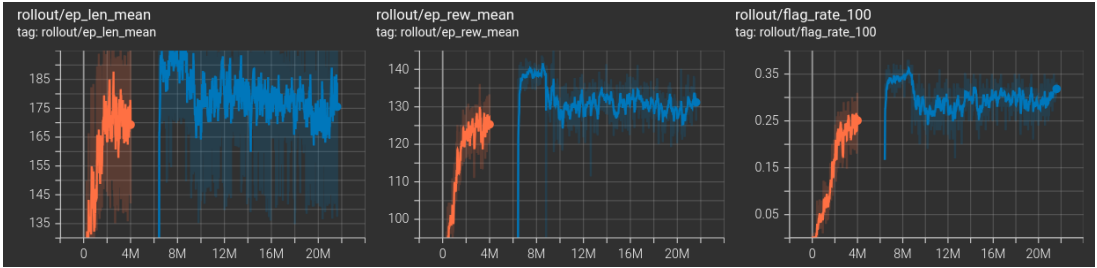
Random start	*how* Mario is placed
Curriculum	*when/how far* to place him

Results: Multi-Task (Worlds 1-1 to 1-4)

Setup: Worlds 1-1 through 1-4, shared PPO policy.

Problem: reward imbalance

1-1 yields rewards earliest ⇒ its gradients dominate ⇒ later worlds stagnate.
Fix: per-task reward normalisation or gradient surgery [6].



Discussion

Key takeaways:

1. Algorithm > representation: same obs., DQN fails, PPO succeeds
2. Symbolic = 6× faster wall-clock, higher peak performance
3. On-policy transfer → catastrophic forgetting
4. Multi-task needs reward balancing

Trade-off

Pixel	General, no domain knowledge
Symbolic	Fast, compact, but fragile

Conclusion

Contributions:

- PPO strongly outperforms DQN on Mario
- Symbolic states: $6\times$ speedup, better reward
- Empirical study of catastrophic forgetting
- Reward imbalance identified as multi-task bottleneck

References I

- J. Schulman et al. *PPO Algorithms*. arXiv:1707.06347, 2017.
- V. Mnih et al. *Human-level control via DRL*. Nature, 518, 2015.
- A. Raffin et al. *Stable-Baselines3*. JMLR, 22(268), 2021.
- M. McCloskey, N. J. Cohen. *Catastrophic Interference*. Psych. Learning & Motivation, 24, 1989.
- J. Kirkpatrick et al. *Overcoming Catastrophic Forgetting*. PNAS, 114(13), 2017.
- T. Yu et al. *Gradient Surgery for MTL*. NeurIPS, 2020.

Thank You

Questions?

Lucas Martim · Sergio Contente · Leonardo Falabella · Gabriel Corsi · Lara Polachini

CSC-52081 · Reinforcement Learning · March 2026